Esteban I. Ramos

estebanramo@gmail.com · esiramos@ucsc.edu · https://eirnf.github.io/ · linkedin.com/in/esteban-ramos-956a34174/ · github.com/EIRNf

Research Interests

Systems research spanning distributed systems, microservices, and novel hardware interconnects. Current PhD work applies the symbolic AI programming model of propagator networks to provide stronger guarantees and explainable behavior for black box AI models. Prior work leveraged SmartNICs and Compute Express Link (CXL) to reduce end-to-end latency in cloud-scale applications, and contributed to mechanized verification tooling for systems software.

Education

University of California, Santa Cruz — Santa Cruz, CA

Ph.D. Student, Computer Science — 2021 – Present
Advisor: Leilani Gilpin

M.S. in Computer Science — Granted December 2025
Advisor: Peter Alvaro
Thesis: *Leveraging Datacenter Hardware to Mitigate Microservice Performance Shortcomings* (see Publications [3])

Emory University — Atlanta, GA

B.S. in Computer Science; Minor in Applied Mathematics — May 2019

Publications

[1] Madeleine Finn, Aditya Singla, Stephen Sheng, **Esteban Ramos**, 2026. **Grounding LLM Decisions with Lexicographically Ordered Constraints for Autonomous Driving**. In *BayLearn - Machine Learning Symposium San Francisco Bay Area 2026*

[2] Aditya Singla, Stephen Sheng, Madeleine Finn, **Esteban Ramos**, 2026. **Adaptable and Transparent Assistance in Appendicitis Diagnosis** . In *BayLearn - Machine Learning Symposium San Francisco Bay Area 2026*

[3] **Esteban I. Ramos**. 2025. **Leveraging Datacenter Hardware to Mitigate Microservice Performance Shortcomings**. M.S. Thesis, University of California, Santa Cruz. Advisor: Peter Alvaro
Committee: Peter Alvaro, Ike Nassi, Andi Quinn
`https://escholarship.org/uc/item/6rz5918g`

[4] Tim Goodwin, **Esteban Ramos**, Andi Quinn, and Lindsey Kuper. 2025. **Monolift: Automating Distribution With the Tools You Have at Home**. In *PLOS '25: Proceedings of the 13th Workshop on Programming Languages and Operating Systems*. ACM.

[5] Peter Alvaro, Matthew Adiletta, David Cheng, Adrian Cockroft, Frank Hady, Ramesh Illikkal, **Esteban Ramos**, and Robert Soulé. 2024. **NotNets: Accelerating Microservices by Bypassing the Network**. In *Proceedings of the 15th ACM SIGOPS Asia-Pacific Workshop on Systems (APSys '24)*. ACM, New York, NY, USA, 67–73.
`https://doi.org/10.1145/3678015.3680494`

[6] **Esteban Ramos**, Robert Soulé, Peter Alvaro, Pietro Bressana, Edmund Chen, Uri Cummings, Rui Li, James Tsai, and Rajit Manohar. 2024. **Split gRPC: An Isolation Architecture for RPC Software Stacks**. In *Proceedings of the 15th ACM SIGOPS Asia-Pacific Workshop on Systems (APSys '24)*. ACM, New York, NY, USA, 81–87.
`https://doi.org/10.1145/3678015.3680484`

[7] Emily Rexer, Ishan Saran, and **Esteban Ramos**. 2019. **A Monetary Evaluation of Ecosystem Services**. *The UMAP Journal* 40(3). *(MCM/ICM Outstanding Winner, selected for publication)*

Experience

University of California, Santa Cruz — Santa Cruz, CA

Graduate Student Researcher, Artificial Intelligence Explainability and Accountability (AIEA) Lab / Language, Systems, and Data Lab
Fall 2021 – Present

- Researching the propagator networks programming model to provide guarantees for applications the leverage non-deterministic black box models. Mentored high school students in two projects that were both accepted into the Baylearn Symposium (see Publications [1,2])

- Researched techniques to reduce end-to-end request latency in microservices by leveraging SmartNICs and Compute Express Link (CXL); resulted in two publications at APSys '24 (see Publications [3, 5, 6])
- Developed an open-source synchronization benchmark for multi-host distributed shared memory environments; paper to be submitted to OSDI '26
- Contributed to DARPA-funded project TWISTED (W912CG-22-C-0011): implemented an end-to-end mechanized verification pipeline for OS development in Rust

Teaching Assistant Spring 2022 – Present

- Led lab sections, review sessions, and office hours; graded assignments, exams, and maintained automated testing infrastructure for: CSE130: Principles of Computer Systems Design, CSE 113: Parallel and Concurrent Programming, CSE 180: Database Systems I, CSE 181: Database Systems II, CSE 182: Introduction to Database Management Systems

Nera Systems Remote, CA
Consultant Summer 2024

- Stealth startup focused on providing third parties shared data execution over secure data by leveraging Fully Homomorphic Encryption, worked as a compiler infrastructure consultant to provide the initial architecture for transpiling existing data processing code into secure equivalents
- Developed a compiler that transforms standard Python source code into functionally equivalent code compliant with Fully Homomorphic Encryption (FHE), enabling execution over private encrypted data
- Established a compilation and execution pipeline with configurable software and FPGA-based backends for the TFHE encryption scheme

Intel Corporation Santa Clara, CA
Graduate Software Engineering Intern Summer 2022

- Developed a prototype offloading RPC processing to Intel's Infrastructure Processing Unit (IPU)
- Achieved zero-copy request/reply flow in a gRPC offload prototype through low-level profiling and latency optimization
- Work contributed to Publication [6] at APSys 2024

NCR Corporation Atlanta, GA
Software Engineer Spring 2020 – Fall 2021

- Engineered cloud-scale microservices applications on an experienced Agile team, deployed on Kubernetes with distributed batch-processing frameworks (Apache Beam, Google Dataflow, Apache Spark)
- Designed and built an automatic schema-flexible ETA ingestion platform for batch and stream processing of customer records

SimBioSys Lab, Emory University Atlanta, GA
Undergraduate Student Researcher Fall 2019

- Researched synthesis of distributed traces of HPC motifs and patterns to benchmark systems on an upscaling simulator

Verint Systems Atlanta, GA
Software Engineering Intern Summer 2019

- Developed a RESTful API in Java for querying internal status information of a complex integration component; delivered as a self-led project that was pushed to production upon completion

Talks

[T1] **Esteban Ramos**. 2024. **NotNets: Accelerating Microservices by Bypassing the Network**. Presented at the *15th ACM SIGOPS Asia-Pacific Workshop on Systems (APSys '24)*, Kyoto, Japan.
`https://doi.org/10.1145/3678015.3680494`

[T2] **Esteban Ramos**. 2024. **Split gRPC: An Isolation Architecture for RPC Software Stacks**. Presented at the *15th ACM SIGOPS Asia-Pacific Workshop on Systems (APSys '24)*, Kyoto, Japan.
`https://doi.org/10.1145/3678015.3680484`

[T3] **Esteban Ramos**. 2023. **Split gRPC: An Isolation Architecture for RPC Software Stacks**. Presented at the *Computational I/O Stack Workshop*, Santa Cruz, CA.
`https://skyhookdm.github.io/event/20230817/`

Awards and Honors

- **Outstanding Winner**, MCM/ICM: The Interdisciplinary Contest in Modeling (COMAP), Spring 2019 — one of 19 Outstanding Winners from 11,000+ participating teams.

Technical Skills

Programming Languages: Proficient in C, C++, Go, SQL, Java; Experience with Rust, Python, JavaScript, MATLAB, x86 Assembly

Frameworks & Tools: Proficient in UNIX/Linux, linux-perf, Kubernetes; Experience with lock-free programming, Kani (Rust mechanized verification), Valgrind, Spring Framework, Apache Beam, Gradle, Maven, Google Cloud Platform

Service and Leadership

Research Mentor, Science Internship Program, University of California, Santa Cruz — Summer 2025,2026

- Mentored high schoolers through the development of two projects implemented with propagator networks programming model which resulted in Publications[1,2].
- Mentored high schoolers through the development of a mutual exclusion benchmark library that resulted in a publication submission.

Raspberry Pi Computer Cluster, Emory University — Fall 2019
Founder Collaborators: Simon Marty, Emily Rexer, Yi Wang

- Received funding from Emory Techlab to design and build a Raspberry Pi cluster providing students a low-cost platform to learn distributed computing and Kubernetes
- Integrated the cluster with the Emory campus network for secure remote access

SC18 (Supercomputing Conference), Student Volunteer — Fall 2018

- Volunteered at the largest annual High-Performance Computing conference

International Students Program, Oxford College of Emory University — Fall 2017 – Spring 2018
International Mentor / Orientation Leader

- Organized International Student Orientation and mentored 12 international students during their first year at Oxford College

Languages

Native Spanish · Fluent English · Proficient in German

References

Available upon request.